Project question

Can we predict which Halifax Transit routes and time periods will have buses passing up waiting riders, and do the university calendar and weather improve that prediction beyond past ridership alone?

Description

When a bus is full, it drives past people waiting at the stop. Halifax Transit logs these as "passenger overload" incidents. In September 2026, HRM released this data publicly for the first time (2020 to present). Overloads are concentrated on a few busy routes, mostly during the afternoon peak, and they rise sharply every September when university classes start. If they can be predicted ahead of time, Halifax Transit could add extra buses where and when they're most needed instead of reacting after riders have already been left behind.

In short: we want to predict whether a bus will pass up riders on a given route, day and time period, using past ridership (APC boardings), the university academic calendar and weather, with the overload incident reports as the record of when it actually happened.

Data

Data	Link	Notes
Overload incidents	HRM Open Data	9,354 incidents, 2020–2026. Times are in UTC, so convert to Halifax time.
APC boardings (current year)	HRM Open Data	Half-hourly boardings by route and direction, 2026.
APC boardings (archived)	Search "Transit Automated Passenger Counts Archived" in the HRM Open Data catalogue	2017 to last year, one CSV per year. Still needed for training.
Weather	Environment Canada historical data	Halifax Stanfield, hourly or daily.
University calendar	Dal Important Dates	Only shows 2026–27. Get 2023–2025 from Dal's archived academic calendars.